

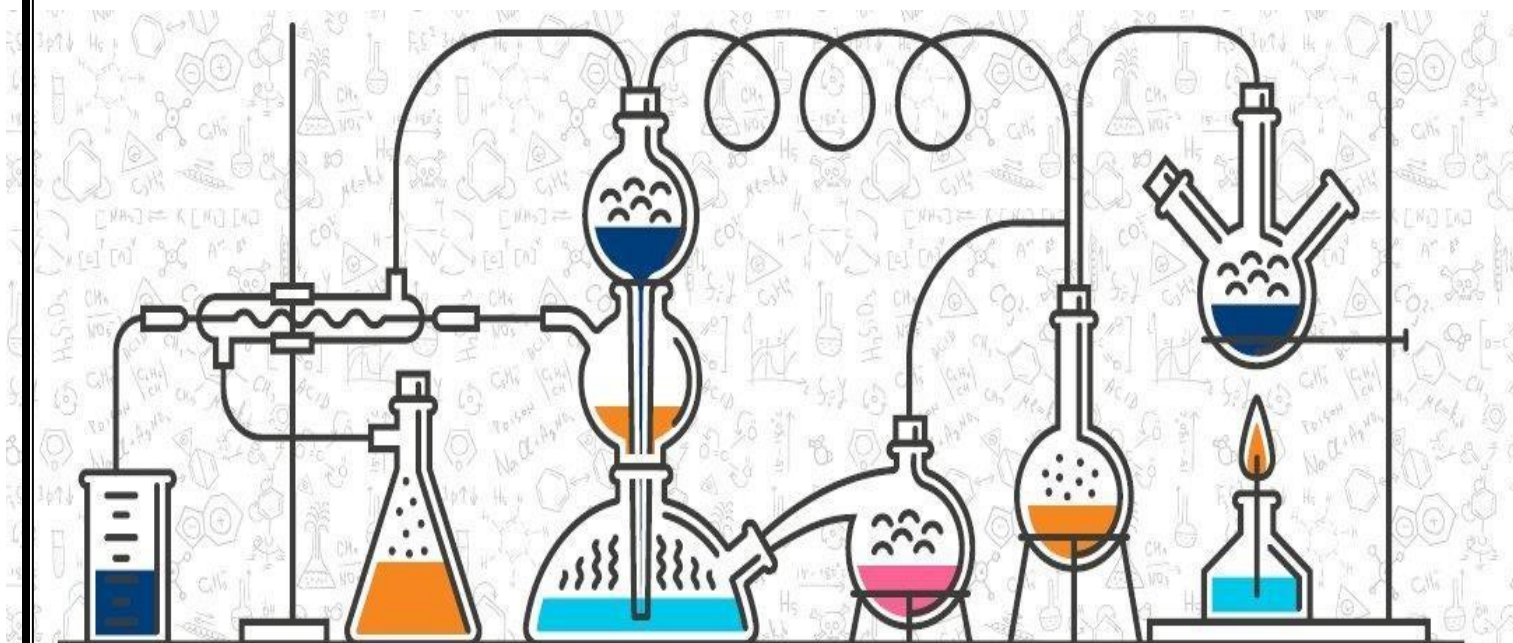


II SEMESTER (1BCHE102/202)

Smart Materials for Emerging Technology - CS Stream

Module – 1

Sustainable Chemistry for Energy Devices



DAYANANDA SAGAR COLLEGE OF ENGINEERING

Accredited by National Assessment & Accreditation Council (NAAC) with 'A' Grade
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi
& ISO 9001:2015 Certified)

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Module-1: Sustainable Chemistry for Energy Devices

FUNDAMENTAL CONCEPTS OF ELECTROCHEMISTRY

A cell is an energy conversion device consisting of two half cells (or single electrodes) namely anode (at which oxidation (loss of electrons) takes place) and cathode (at which reduction (gain of electrons) takes place).

Nernst equation

$$E = E^{\circ} + \frac{2.303 RT}{nF} \log [M^{n+}]$$

Where R= Universal gas constant. T = Absolute temperature. n = number of moles of electrons involved in the reaction. F = Faraday's constant. E° = Standard Electrode potential E = Electrode potential

On substituting the values of R=8.314 J/K/mol, T=298 K, F=96500 C, the above equation reduces to

$$E = E^{\circ} + \frac{0.0591}{n} \log [M^{n+}]$$

ELECTROCHEMICAL CELLS: "These are the devices which convert chemical energy in to electrical and vice versa".

Electrochemical cells are mainly divided into (i) Galvanic cells (ii) Electrolytic cells. **Galvanic cells:**

The devices that convert chemical energy into electric energy are called galvanic cells. **Electrolytic**

cells: The devices that bring chemical change by the expense of electrical energy are called electrolytic cells.

Single Electrode Potential: It is defined as the potential that exists between the metal and its ions in the solution when they are in equilibrium.

Standard Electrode Potential: It is defined as the potential that exists between the metal and its unimolar solution at 298K.

Electromotive force (emf): The potential difference that causes the electrons to flow from one half cell to other half cell is called electro motive force.

The potential difference between two electrodes of a galvanic cell which causes the flow of current from anode (higher potential) to cathode (lower potential) is called the E.M.F.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = E_{\text{Right}} - E_{\text{Left}}$$

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}}$$

Concentration Cells: The cells in which the EMF produced is due to the difference in the concentrations of the electrodes or electrolytes.

There are two types of concentration cells.

i) Electrode concentration cells: A cell is formed by combination of same electrodes of different concentrations dipped in same electrolyte solutions is called electrode concentration cell.

For example: Pt, H₂ (1atm) / H⁺(1M) // H⁺(1M) /H₂(2 atm),Pt

ii) Electrolyte concentration cells: A cell is formed by combination of same electrodes dipped in same electrolyte solutions of different concentration is called electrolyte concentration cell.

For example: $\text{Cu(s)} / \text{Cu}^{2+} (\text{C}_1) // \text{Cu}^{2+} (\text{C}_2) / \text{Cu(s)}$ Where $\text{C}_2 > \text{C}_1$

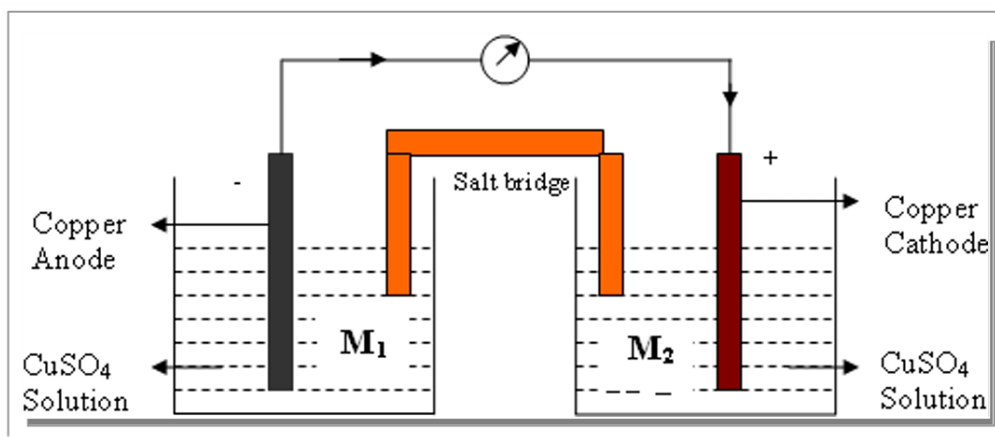
EMF of a concentration cell: Consider a concentration cell

$$\text{M} / \text{M}^{n+} (\text{C}_1) // \text{M}^{n+} (\text{C}_2) / \text{M} \quad \text{Where } \text{C}_2 > \text{C}_1$$

$$E_{\text{cell}} = E_{\text{Cathode}} - E_{\text{anode}}$$

ELECTROLYTE CONCENTRATION CELLS

It is a galvanic cell, which consists of two identical electrodes that are in contact with the same solution of electrolyte at different concentrations. A typical example of copper concentration cell is shown below, It consists of two Copper electrodes immersed in two different concentrations of CuSO_4 solutions. These two electrodes are externally connected by metallic wire and internally by a salt bridge.



Cell representation: The cell can be represented as, $\text{Cu} | \text{Cu}^{2+} (\text{C}_1) // \text{Cu}^{2+} (\text{C}_2) | \text{Cu}$. By convention left-hand electrode is the anode and right-hand electrode is a cathode.

At Anode: $\text{Cu} \rightarrow \text{Cu}^{2+} (\text{C}_1) + 2\text{e}^-$

At Cathode: $\text{Cu}^{2+} (\text{C}_2) + 2\text{e}^- \rightarrow \text{Cu}$

Net reaction: $\text{Cu}^{2+} (\text{C}_2) \rightleftharpoons \text{Cu}^{2+} (\text{C}_1)$

EMF of Concentration Cell

$$E_{\text{cell}} = E_{\text{Mn+}/\text{M}}^0 + \frac{0.0591}{n} \log[\text{C}_2] - E_{\text{Mn+}/\text{M}}^0 - \frac{0.0591}{n} \log[\text{C}_1]$$

$$E_{\text{cell}} = \frac{0.0591}{n} \log[\text{C}_2] - \frac{0.0591}{n} \log[\text{C}_1]$$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \frac{[\text{C}_2]}{[\text{C}_1]}$$

This is the expression of Nernst equation for concentration cell at 25°C

Numerical Problems:

1. A concentration cell was constructed by immersing two silver electrodes in 0.05M and 1M AgNO_3 solution. Write the cell representation and calculate E_{cell} at 298K.

Ans: $\text{Ag}/\text{AgNO}_3 (0.05\text{M}) // \text{AgNO}_3 (1\text{M}) / \text{Ag}$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \frac{\text{C}_2}{\text{C}_1}$$

$$= \frac{0.0591}{1} \log \frac{1}{0.05}$$
$$= 0.0768 \text{ V}$$

2. Represent the cell formed by coupling two Cu-electrodes immersed in CuSO₄ solutions concentrations of Cu²⁺ ions in one electrode system and 100 times more concentrated than other. Calculate the E_{cell} at 298 K.

Ans: Cu/ Cu²⁺ (1M) // Cu²⁺ (100M) / Cu

$$E_{\text{cell}} = \frac{0.0591}{n} \log \frac{C_2}{C_1}$$
$$= \frac{0.0591}{2} \log \frac{100}{1}$$
$$= 0.0298 \times 2$$
$$= 0.0591 \text{ V}$$

Battery:

A Battery is a combination of two or more cells arranged in series or parallel. A battery is a device capable of converting the chemical energy, contained in the active materials that compose it, into electric energy by electrochemical redox reactions

Classification of batteries

1. Primary batteries

1. In primary batteries, the chemical energy is converted into electrical energy as long as the chemical components are active.
2. These batteries the reaction occurs only once and after that they must be discarded.
3. These batteries cannot be recharged as the chemical reactions which occur within the primary batteries are irreversible.

Examples: Zn- air battery, Dry cell, Li- MnO₂

2. Secondary batteries

1. Secondary batteries are those which after discharging can be recharged.
2. These batteries chemical reactions taking place are reversible
3. The redox reaction which converts chemical energy into electrical energy can be reversed by passage of current.
4. The electric energy is stored in the form of chemical energy these batteries are also known a storage cell.

Example: Lead storage battery, Nickel-cadmium battery, Nickel-metal hydride battery and Lithium-ion battery.

3. Reserve Batteries

1. These batteries one of the components is isolated and incorporated into the battery when required.
2. The electrolyte is the component that is isolated, but some water activated batteries contain the electrolyte solute and water is added for activation.
3. They are used to deliver high power for relatively short periods of time in application such as

radiosondes (air borne instruments to send to meteorological information back to earth by radio).
Examples: Mg batteries activated by water (Mg-AgCl, Mg-CuCl), Zn-Ag₂O batteries etc

Li-ion Battery

Construction

It consists of a lithium metal oxide for the positive electrode and lithiated carbon for the negative electrode. These materials are adhered to a metal foil current collector with a binder, typically polyvinylidene fluoride (PVDF) or the copolymer polyvinylidene fluoride hexafluoropropylene (PVDF-HFP), and a conductive diluent, graphite. The positive and negative electrodes are electrically isolated by a microporous polyolefine/polypropylene separator. A solution of LiPF₆ in organic solvent like dimethyl carbonate and ethylene carbonate acts as electrolyte.

Components:

Anode (-ve electrode): Lithiated carbon

Cathode (+ve electrode): Lithiated metal oxide LiMO₂ (M=Co, Mn, Ni)

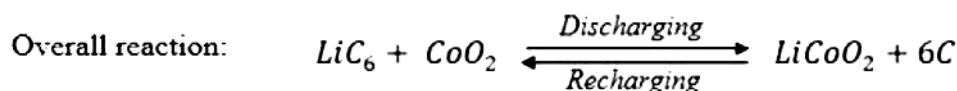
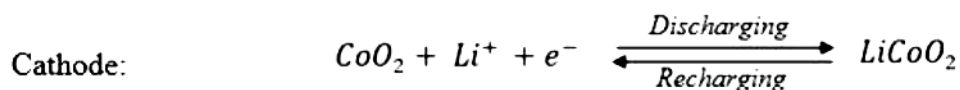
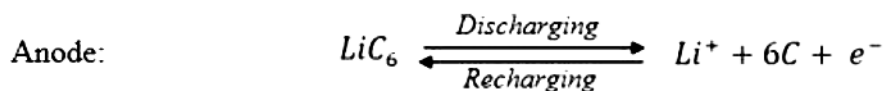
Electrolyte: LiPF₆ in dimethyl carbonate and ethylene carbonate.

Separator: poly olefin/polypropylene.

Li is used as anode because a) Extremely Low Standard Reduction Potential, b) very high specific capacity, High energy density, high energy output.

Working

1. The anode and cathode store the lithium. During discharging Li atoms present in the graphite layer are oxidized, liberating Li ions and electrons. Li ions migrate from anode to cathode through electrolyte through the separator which selectively allows only the Li ion to pass through and block the flow of electrons inside battery.
2. At cathode, Li ions are reduced to Li atoms and are inserted in to the structures of metal oxide.
3. During charging of the battery, Li atoms present in the structure of metal oxide get oxidized liberating Li ions and electrons. The liberated Li ion moves through the separator towards the graphite carbonate electrode (anode).
4. At the graphite electrode, Li ions get reduced to Li atoms and gets inserted into the layered structure of graphite.



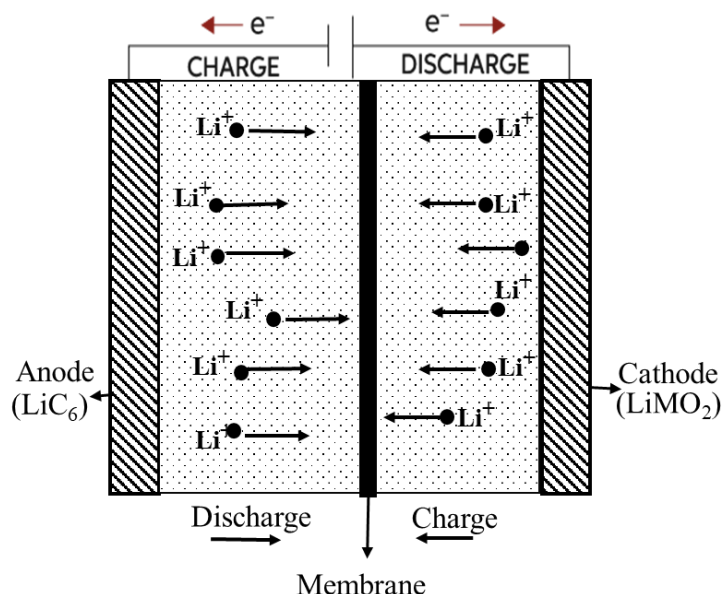


Fig: Li-ion battery

Applications:

It is used in cell phones, laptops, automobiles, defense and aerospace applications.

Li-ion battery has more advantages than traditional rechargeable battery:

- i) High voltage: Lithium batteries have voltages up to about 4 V, depending on the cathode material,
- ii) highly stable
- iii) wide operating temperature (ie. from -40°C to +70°C)
- iv) Flat discharge Characteristics
- v) High specific energy and energy density.

Next-Generation Energy Systems:

Next-generation energy systems integrate advanced storage, smart grids, and renewable generation to deliver high efficiency, low carbon emissions, and enhanced reliability. While technologies such as Solid-state battery and hydrogen fuel cells support high energy density and clean fuel conversion, large-scale grid stability requires durable and scalable storage solutions.

Redox flow battery

Introduction

Redox Flow Batteries (RFBs) are electrochemical energy storage systems in which energy is stored in liquid electrolytes containing redox-active species. They are especially suited for grid-scale and long-duration storage.

Vanadium redox flow batteries (VRFBs)

Vanadium redox flow batteries (VRFBs) is one of the example of redox flow battery, which store energy through reversible chemical reactions in liquid electrolytes, offering scalable power for grid applications. It uses vanadium ions in different oxidation states for charging and discharging.

Construction

VRFBs feature two external tanks containing electrolytes: the **Anolyte** (V²⁺/V³⁺ in sulfuric acid) and **Catholyte** (V⁴⁺/V⁵⁺ in sulfuric acid). These tanks connect via pumps to a central electrochemical cell

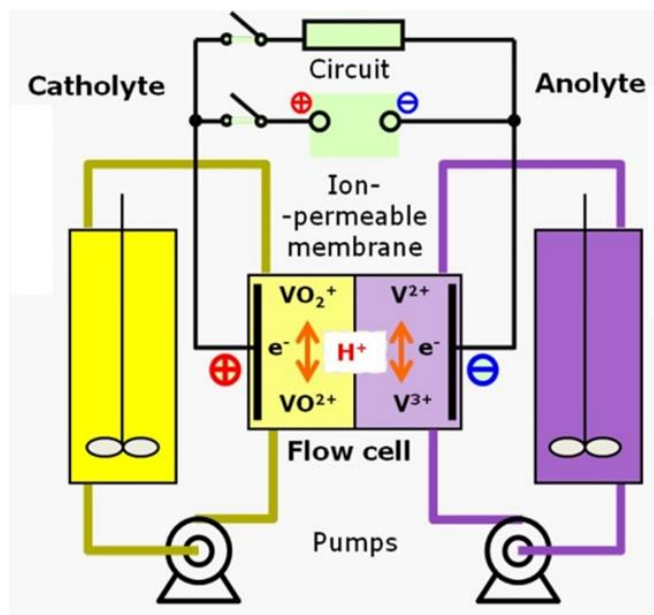
with a negative electrode, positive electrode (often carbon felt), and an ion-exchange membrane that allows proton (H^+) transport while preventing vanadium crossover.

Electrodes: Carbon felt (porous, 80% void) (Reaction sites, electron transfer)

Membrane: Nafion (perfluorosulfonic acid) (H^+ conduction, crossover block)

Negative electrolyte (Anolyte): $V_2(SO_4)_3$ in sulphuric acid

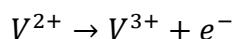
Positive electrolyte (Catholyte): VO_2SO_4 in sulphuric acid



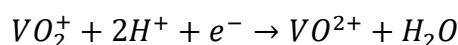
Working

During Discharge (Battery Produces Power)

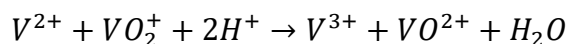
Negative electrode (oxidation):



Positive electrode (reduction):

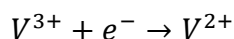


Overall reaction

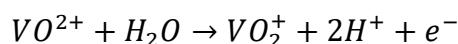


During Charging (Energy Storage)

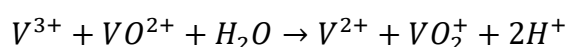
Negative Electrode (Reduction)



Positive Electrode (Oxidation)



Overall Charging Reaction



Redox Flow Battery for EV Applications

- A redox flow battery stores electrical energy in liquid electrolytes kept in external tanks, and electricity is generated through redox reactions in a cell stack.
- It provides high safety because the electrolytes are non-flammable, reducing the risk of fire in electric vehicles.
- Redox flow batteries exhibit a long cycle life with stable performance over many charge–discharge cycles.
- They offer the potential for rapid refueling by replacing or refilling the electrolyte rather than recharging the battery.
- Due to their high durability and safety, redox flow batteries are suitable for heavy-duty EVs, buses, and fleet vehicles, where space and weight constraints are less critical.

Supercapacitor:

A supercapacitor is an electrochemical energy storage device that stores energy by separating ions at the interface between an electrode and an electrolyte. It is bridging the gap between the conventional capacitors and batteries. They are capable of providing higher energy with higher power density.

Ultra-small asymmetric supercapacitor

Ultra-small asymmetric supercapacitors are tiny devices designed for low-power, miniaturized electronics such as IoT nodes and wearables.

Asymmetric supercapacitors store the charge by combination of two different mechanisms. Where one electrode, usually the positive electrode, will be made of pseudo-capacitors and the negative electrode made of an electrochemical double layer, integrating high-performance energy storage into limited spaces, such as wearable electronics, medical implants, micro-sensors and flexible devices.

Construction and working of Ultra-small asymmetric super capacitor:

An ultra-small asymmetric supercapacitor uses two distinct electrode materials—one for electrostatic energy storage (like activated carbon) and another for faradaic reactions (like metal oxides)—separated by a separator and filled with an electrolyte. Construction involves sandwiching the separator between the two different electrodes and immersing them in the electrolyte, all within a compact device suitable for size-constrained applications.

Anode: It is electrochemical double layer (EDLC)- activated carbon or graphene, (commonly used for Metal ion adsorption)

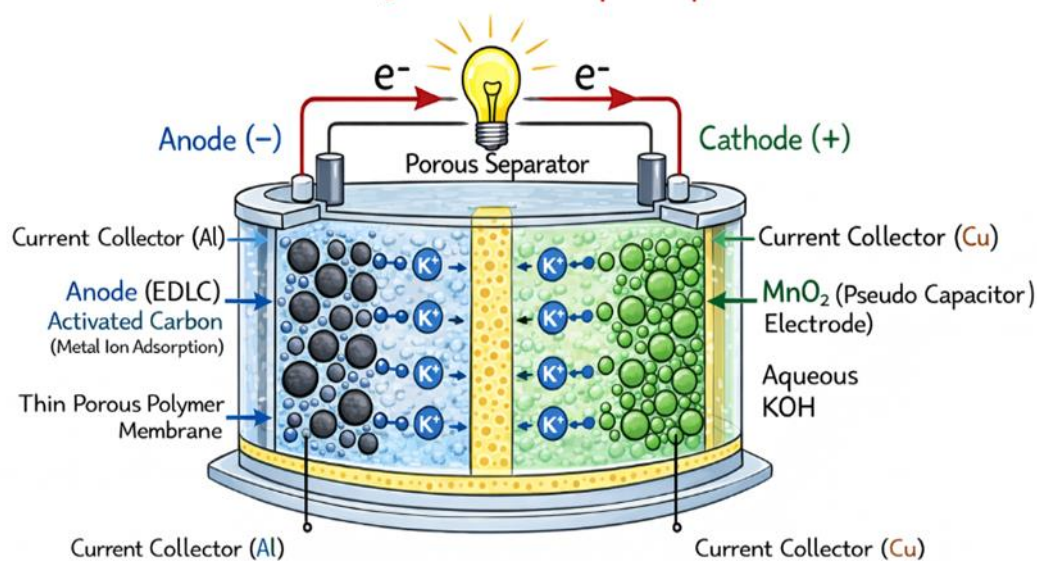
Cathode- A Pseudo capacitor electrode, MnO_2 or RuO_2

Separator: A thin porous polymer membrane

Electrolyte: Aqueous solution of KOH.

Current Collectors: Thin, flexible, conductive metal foils (like Al or Cu)

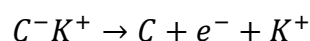
Ultra-small Asymmetric Supercapacitor



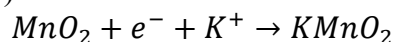
Working:

Discharging Process

Anodic reaction (Negative electrode)



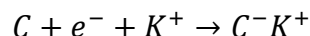
Cathodic reaction (Positive electrode)



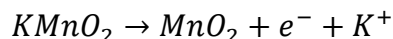
Overall Discharge Reaction: $C^{-}K^{+} + MnO_2 \rightarrow C + KMnO_2$

Charging Process

Cathodic reaction (Negative electrode)



Anodic reaction (Positive electrode)



Overall Charging Reaction: $C + KMnO_2 \rightarrow C^{-}K^{+} + MnO_2$

Applications of Ultra-Small Asymmetric Super capacitors

1. Wearable Electronics

- Ultra-small asymmetric supercapacitors are compact, lightweight, fast charging, high power density, flexibility, long cycle life, and safe solid-state operation, and flexible energy storage devices suitable for wearable electronics.
- Their main applications include:
- Smart watches and fitness bands – Provide quick charging and support short bursts of power for displays and wireless communication.

- Health monitoring patches – Power ECG, glucose, and temperature sensors with stable output.
- Smart textiles (e-textiles) – Embedded into clothing for energy storage in flexible fabrics.
- Wireless body sensors (IoT devices) – Supply power for data transmission in body area networks.
- Medical implants – Offer reliable, long cycle life energy storage for low-power bioelectronics.

2.IoT: IoT devices that can benefit from ultra-smart asymmetric supercapacitors:

- Smart home sensors (temperature, humidity, motion detectors)
- Environmental sensors (air quality monitors, soil moisture sensors)
- Industrial IoT sensors (vibration, pressure, and gas detectors)
- Smart agriculture devices (crop monitoring, irrigation controllers)
- Asset tracking tags (RFID and GPS trackers)
- Smart watches and e-textiles
- Wireless sensor nodes in remote or hard-to-access locations

Clean Energy

Clean energy refers to energy generated from renewable and environmentally friendly sources that produce little or no greenhouse gas emissions. It includes solar, wind, hydro, geothermal, and sustainable bioenergy, helping reduce pollution and support environmental sustainability.

Fuel cell:

A fuel cell is an electrochemical device that converts chemical energy from hydrogen and oxygen into electricity, heat, and water without combustion. It is an important clean energy technology that provides efficient and low-emission power for transportation, backup systems, and stationary applications.

The **core components** of a fuel cell are:

- **Anode** – where the fuel (e.g., hydrogen) is oxidized, releasing electrons and ions.
- **Cathode** – where the oxidant (e.g., oxygen) is reduced, consuming electrons and ions.
- **Electrolyte** – the medium that allows ionic conduction between the anode and cathode while preventing the direct passage of electrons.

Difference between fuel cells and Battery

Fuel Cells	Batteries
Do not store chemical energy	Stores chemical energy
Reactants are fed from outside continuously	The reactants form an integral part of it
Need expensive noble metal catalysts	Not required
No need of charging	Gets discharged when stored-energy is exhausted
Never become dead	Limited life span in use
Useful for long-term services for electricity generation	Useful as portable power

Solid Oxide Fuel Cells (SOFC)

A Solid Oxide Fuel Cell (SOFC) is a high-temperature electrochemical device that uses a solid ceramic electrolyte instead of a liquid. It typically operates in the range of 800–1000°C (1470–1830 F), which enables high efficiency and the direct use of a variety of fuels.

Construction of Solid Oxide Fuel Cell (SOFC)

An SOFC is built in a layered structure with three essential components anode, electrolyte, and cathode.

1. Electrolyte

- Material: Solid ceramic oxide, most commonly yttria-stabilized zirconia (YSZ).

2. Anode

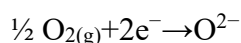
- Material: Porous nickel/YSZ cermet (ceramic–metal composite).

3. Cathode

- Material: Strontium-doped lanthanum manganite (LSM) or related perovskite oxides.

Working Principle

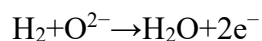
Cathode (Reduction of Oxygen): Oxygen from air is supplied at the cathode, where it is reduced by electrons arriving through the external circuit. The resulting **oxide ions (O²⁻)** migrate through the solid electrolyte crystal lattice toward the anode.



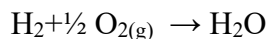
Electrolyte (Ion Conductor): The stabilized zirconia ceramic allows the transport of **O²⁻ ions** while blocking electron flow, maintaining charge separation.

2. **Anode (Oxidation of Hydrogen):** When fuel gas (hydrogen or carbon monoxide) is supplied to the anode, the arriving O²⁻ ions oxidize the fuel:

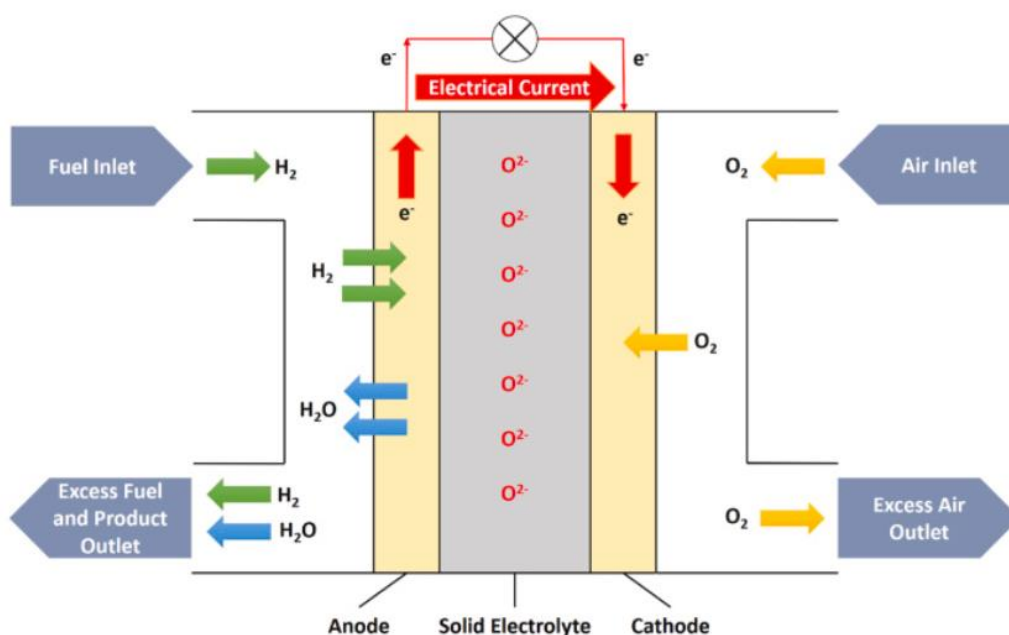
For hydrogen as a fuel,



The released electrons return to the cathode through an external circuit, generating **direct current (DC) electricity**. Overall, Cell Reactions, with H₂ as fuel:



SOFC conducts **oxide ions** from cathode to anode. Electrons travel through the external circuit, producing **DC electricity**.



Applications

1. Stationary power generation (Supplying electricity directly at or near the point of use)
2. Industrial Applications
3. Transportation
4. Military Applications
5. Residential and Commercial Power

Limitations:

1. High operating temperature (600–1000 °C) causes long start-up times.
2. Material challenges: thermal stress, electrode/electrolyte degradation.
3. Expensive ceramic materials and fabrication costs.
4. Limited portability (better for stationary applications).
5. Requires pure fuel or reforming units.

Photovoltaic cells

Photovoltaic cells or Solar cells are the semiconductor devices which convert sunlight into direct current electricity on illumination. The radiations reaching earth from the sun and converting them into different useful forms of energy is called solar energy.

The utilization of solar energy is of two types – Direct solar power and indirect solar power.

Principle

When sunlight (photons) falls on the surface of the solar cell: The energy of each photon is given by Planck's equation:

$$E = h\nu = hc/\lambda$$

Construction of a Solar Cell

1. **Semiconductor Base Material:** A solar cell is primarily made of silicon, a semiconductor that can absorb light and generate charge carriers.
2. **Formation of p–n Junction:**
 - a. The top layer is phosphorus-doped silicon (n-type), which has extra electrons as majority carriers.
 - b. The bottom layer is boron-doped silicon (p-type), which has “holes” (positive charge carriers) as majority carriers.
 - c. When these two layers are joined, a p–n junction diode is formed at their interface.
3. **Front Contact (Metallic Grid):** A fine metallic grid is deposited on the top surface. It acts as an electrical contact while still allowing sunlight to pass through the gaps.
4. **Antireflective Coating:** A thin coating is applied between grid lines to minimize reflection losses and enhance light absorption.
5. **Back Contact:** The rear surface of the cell is coated with a metallic layer to complete the circuit and act as the second electrical contact.

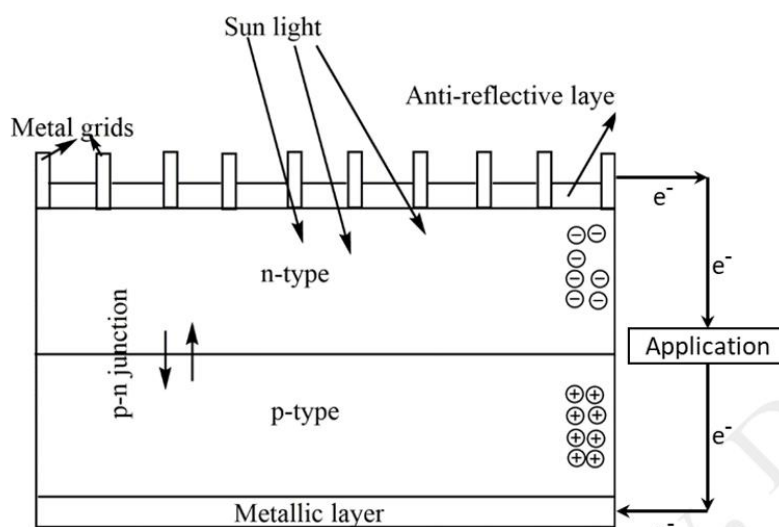


Fig: Si Solar cell

Working

1. **Absorption of Light:** The energy from incident photons is absorbed by the semiconductor material (silicon). The energy of incoming solar radiation is carried by photons.
2. **Generation of Charge Carriers:** The absorbed energy excites electrons from the valence band to the conduction band, leaving behind holes in the valence band. This creates electron–hole pairs.
3. **Separation of Charges (at the p–n junction):** Due to the electric field at the p–n junction, electrons are pushed toward the n-type layer, and holes are pushed toward the p-type layer. This prevents recombination and ensures separation of charges.
4. **Flow of Current:** When an external circuit is connected between the n-type and p-type ends, the electrons flow from the n-side through the circuit to the p-side. This movement of electrons constitutes a direct current (DC). A single PV cell generates a voltage of about **0.5–0.6 V** under standard conditions. To obtain higher voltage and power, many cells are connected in **series (for voltage)** and **parallel (for current)** to form **solar panels**.

Applications:

- ❖ **Domestic Use** – Provides electricity for homes (lighting, appliances, rooftop systems).
- ❖ **Industrial & Power Generation** – Used in factories and large solar power plants to supply grid electricity.
- ❖ **Rural Electrification** – Supplies power to remote areas for lighting, water pumping, and basic needs.
- ❖ **Portable & Small Devices** – Powers calculators, watches, and mobile chargers.
- ❖ **Special Applications** – Used in satellites, street lights, traffic signals, and emergency backup systems

Limitations

- Low efficiency
- High installation cost
- Intermittent power generation
- Large area requirement
- Energy storage issue

Green Hydrogen

Green hydrogen is a **clean energy carrier** produced through the **electrolysis of water** using electricity from renewable sources such as solar and wind power. This **carbon-neutral process** makes it a vital solution for decarbonizing hard-to-abate sectors like transport, shipping, and heavy industry by replacing fossil fuels.

Key Benefits:

- Supports **energy independence** by reducing reliance on imported fossil fuels.
- Contributes to achieving **net-zero emission targets**.
- Provides a sustainable pathway for large-scale clean energy storage and distribution.

Production of green hydrogen by Photocatalytic water splitting (TiO₂) method

Photocatalytic water splitting is an **artificial photosynthesis process** in which water is split into **hydrogen and oxygen** using a photocatalyst and natural sunlight. This process provides a sustainable pathway for clean hydrogen production.

Role of Catalyst:

- The catalyst plays a crucial role in absorbing light, generating charge carriers, and facilitating redox reactions.

- **Titanium dioxide (TiO₂)** is the most widely used photocatalyst due to its suitable band structure and stability.
- Co-catalysts such as **platinum (Pt)** or **palladium (Pd)** are often added to enhance hydrogen evolution efficiency.

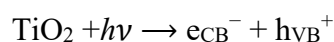
Particulate Photocatalysis (PC) Systems:

- The particulate photocatalyst is dispersed in water and activated under sunlight.
- Upon excitation, the photocatalyst generates electrons and holes that drive the splitting of water molecules.
- Example: TiO₂ with Pt as a co-catalyst for enhanced H₂ production.

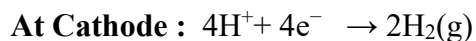
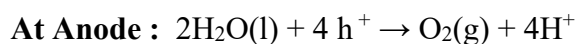
Reaction Mechanism:

Photocatalytic water splitting generally occurs in three steps:

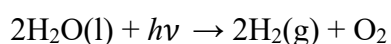
1. **Light absorption** by the semiconductor photocatalyst, generating electron (e⁻)–hole (h⁺) pairs.



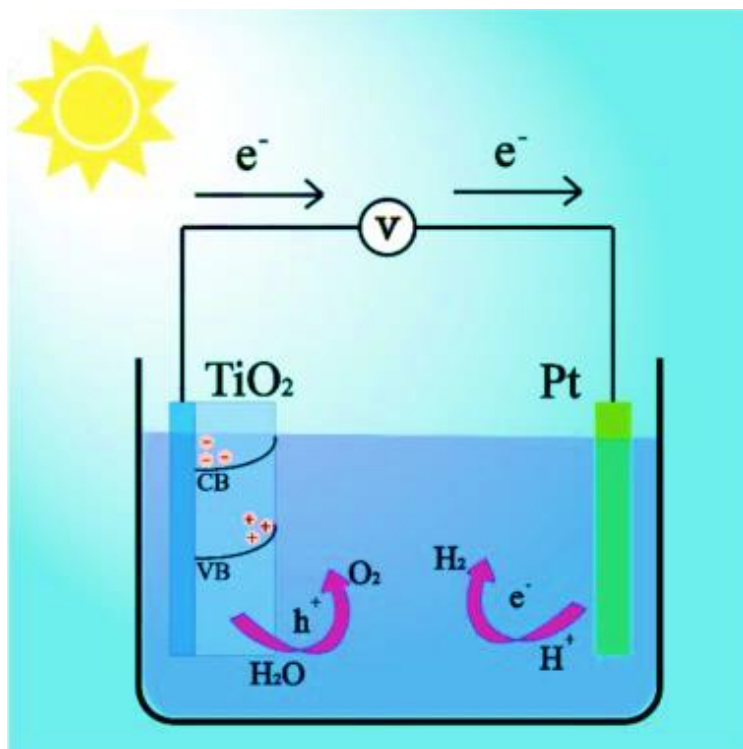
2. **Charge separation and migration** of e⁻ and h⁺ to the catalyst surface.



Overall Reaction



3. **Surface reactions:** Co-catalysts (Pt, Ni, Ru) are often deposited on TiO₂ to enhance H₂ production and suppress electron–hole recombination.



Advantages of TiO₂ Photocatalyst

- Abundant & Low-Cost: Titanium is widely available, cheap, and non-toxic.
- High Chemical Stability: Resistant to corrosion under UV and aqueous conditions.
- Environmentally Friendly: No CO₂ or harmful byproducts.
- Reusable: Can be recycled over many cycles.
- Tunable Properties: Doping (N, Ag, Ir) or heterojunction formation (CdS, graphene) can extend absorption into visible light and improve efficiency.

AI assisted solar panel orientation and efficiency predictions tools:

AI-assisted tools for solar panel orientation and efficiency predictions optimize energy generation by analyzing environmental, weather, and structural data to determine the best tilt and orientation. These tools can increase energy output by 15%–30% compared to traditional methods by utilizing machine learning for real-time tracking, shading analysis, and predictive maintenance.

AI in Panel Orientation

Dynamic Real-Time Tracking: Unlike traditional sun trackers that rely on simple sensors, AI-driven systems use machine learning to predict the sun's position and proactively adjust panels in advance, ensuring optimal positioning even during inconsistent weather.

Adaptive Shading Management: AI detects shadows cast by trees, buildings, or other panels and automatically

adjusts the orientation of affected panels to minimize energy loss.

Automated Layout and Design: AI-powered design tools (like EasySolar or CAD-integrated AI) use satellite imagery and LiDAR to analyze roof geometry, automatically determining the best placement, tilt, and spacing to achieve maximum capacity.

Intelligent Cleaning & Safety: AI controls robotic cleaners and drones, using real-time orientation data to align cleaning mechanisms, ensuring complete coverage without collision.

Efficiency Prediction Tools

Easy Solar AI: Specializes in site and shade analysis using aerial imagery, providing up to 95% accuracy in minutes. It evaluates thousands of layout configurations to improve yields by 3–8%.

Heliolytics: Uses AI-powered drone imagery analysis to detect panel defects (like micro-cracks or shading) and improve system output by 1–3%.

Smart Helio Autopilot: Monitors inverters and sensors in real-time, predicting failures before they impact performance.

SolarEdge Designer: A tool that leverages AI to recommend optimal panel configurations based on site-specific shading, location, and roof angles.

Benefits of AI- assisted tools for solar panel orientation:

1. Enhanced energy yield and efficiency
2. Faster design time and accurate site selection
3. Reduced operational and maintenance costs
4. Accurate forecasting and load balancing

Question Bank :

4 mark questions

1. What is a concentration cell? Explain its types with examples.
2. Calculate the EMF of a silver concentration cell when the concentrations are 0.05 M and 1 M at 298K.
3. A concentration cell is formed using Ag/Ag^+ (0.01 M) and Ag/Ag^+ (1 M). Calculate EMF at 298 K.
4. Write the advantages and applications of Li-ion batteries over conventional batteries.
5. What is a Redox Flow Battery (RFB)? Mention its main components. Write the advantages of redox flow batteries for EV applications.
6. Define supercapacitor. Classify its types.
7. What is an ultra-small asymmetric supercapacitor? Mention its components.
8. Write applications of ultra-small asymmetric supercapacitors in IoT devices.
9. Define fuel cell. Write its components.

10. Difference between fuel cell and battery.
11. Write advantages and limitations of SOFC.
12. List advantages and limitations of solar PV cells.
13. Define green hydrogen. Why is it important?
14. List the role of TiO_2 in photocatalytic water splitting.
15. What is AI-assisted solar panel orientation? Mention its benefits.

6/8 marks

1. Define concentration cell. Derive the expression for EMF of an electrolyte concentration cell using the Nernst equation.
2. Explain classification of batteries in detail with suitable examples.
3. Describe in detail the construction, working, and applications of a Li-ion battery with neat diagram and reactions.
4. Explain construction and working of redox flow battery with diagram. Discuss its advantages for EV and grid storage applications.
5. Describe construction and working of ultra-small asymmetric supercapacitor with neat diagram and reactions.
6. Mention the applications of ultra-small asymmetric supercapacitors for wearable devices and IoT devices.
7. Explain construction and working principle of Solid Oxide Fuel Cell with reactions with diagram and reactions.
8. Explain construction and working of solar photovoltaic cell with neat diagram and its advantages.
9. Explain photocatalytic water splitting using TiO_2 for green hydrogen production with mechanism with diagram and reactions.
10. Discuss the role of AI in improving solar panel orientation and its efficiency.